

A method to Enhance Power Amplifier Linearity using Time delay estimation & correction Technique

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Abstract: Power amplifier, a critical component of the RF chain, is the main source that is responsible for signal distortion. Its nonlinear behavior must be taken into consideration for efficient design to cater the need for high data links. In this paper, techniques for mitigation of distortion for catering needs of the modern era of communication system via feasible digital pre-distortion method have been implemented. Memory effect parameters of the Power amplifiers are calculated with time delay estimation & correction technique such that it behaves as a linear system. These parameters are calculated using the cross-correlation technique.

Keywords— Power amplifier (PA), Error vector magnitude (EVM), Digital Pre-Distortion (DPD), OFDMA, Pre-distortion Least Mean Square (LMS), adjacent channel power ratio (ACPR).

I. INTRODUCTION

With the fast-changing environment and need for high-speed data transfer for information, the requirements of data rates for wireless communication systems are increasing monotonically. This can be feasible by using large bandwidth and more advanced communication techniques, such as MIMO, QPSK, and OFDMA. At the same time, requirements for RF components in data links such as power amplifiers, digital transceivers, mixers, and signal sources, etc. are becoming more stringent so that signal quality should not be greatly degraded by the transmitter. The role of mobile communications [1] networks in military operations continues to grow in importance, with mission areas such as covert special operations, time-critical targeting, command, and control, reconnaissance, surveillance, electronic warfare systems require higher data rates transmission in order to meet the higher demand of quality services [2].

The main component which introduces non-linearity in the transmission link is high power amplifiers (HPAs). Power amplifier non-linearities cause unwanted spectral spreading, in-band distortion, and constellation warping. In wideband systems, the power amplifier may have a memory element, which is caused by the reactive elements of the amplifier.

Several benefits may be obtained by compensating distortions from nonlinear components. Better energy efficiency is obtained with a given nonlinear component and, more generally, for the whole system. Because of out-of-band distortions (spectral regrowth), there is a need to increase guard bands between users and systems.

II. CONCEPTUAL DESIGN

A. Time delay estimation & correction

When data is collected from different sensors, there is some time delay because of path differences or sensor internal architecture. In the case of DPD, PA output and input are used to find the coefficient of an inverse function. Due to PA memory architecture, there has been some delay in response to PA. To find the time difference between input and response of PA there are several techniques. One of the robust techniques which can work in a noisy condition is discussed in the paper. For continuous functions $f(t)$ and $g(t)$ the correlation is defined as

$$[f^*g](\vartheta) = \int_{-\infty}^{\infty} f(t)g(t + \vartheta)dt \quad (1)$$

ϑ is displacement or lag and $[f^*g]$ is the magnitude of cross correlation function. Equation (1) can also be written as

$$[f^*g](\vartheta) = \int_{-\infty}^{\infty} f(t - \vartheta)g(t)dt \quad (2)$$

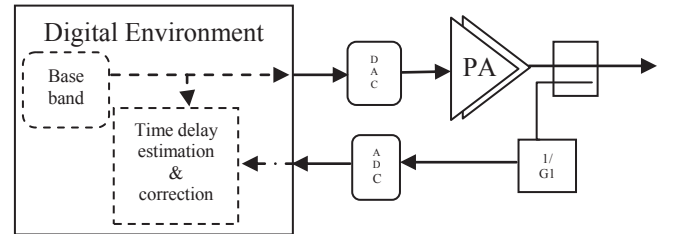


Fig. 1. Time delay estimation & correction technique

The time delay estimation & correction is being done in the digital (discrete) domain. The above two real-valued functions $x(n)$ and $y(n)$ are same but only differing by an unknown shift along the x-axis. Cross-correlation of discrete signal $x(n)$ and $y(n)$ is

$$[xy](\vartheta) \triangleq \sum_{n=-\infty}^{\infty} x(n) * y(n - \vartheta) \quad (3)$$

$[xy](\vartheta)$ is maximum when both $x(n)$ and $y(n)$ are similar or the same in the time domain.

$$Z(\vartheta_1) = \max \{[xy](\vartheta)\} \quad (4)$$

So ϑ_1 is the delay or advance parameter corresponding to $x(n)$ and $y(n)$ cross-correlation magnitude is maximum.

Correction to the delay for DPD algorithm

$$Xcorr(n) = x(1:end - \vartheta_1) \quad (5)$$

$$Ycorr(n) = y(1 + \vartheta_1 : end) \quad (6)$$

Data captured in a digital environment in lab setup is plotted with equation (3)

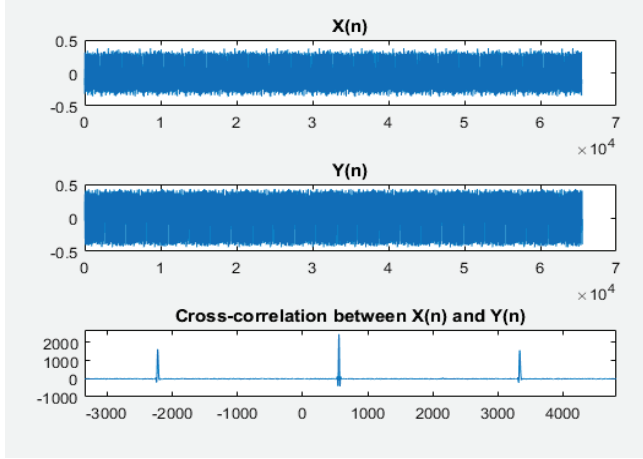


Fig. 2. Delay estimation between $x(n)$ and $y(n)$

τ is calculated as 559. Correction to delay with equation (5) and (6)

$$X_{corr}(n) = x(1:end-559), Y_{corr}(n) = y(1+559:end) \quad (7)$$

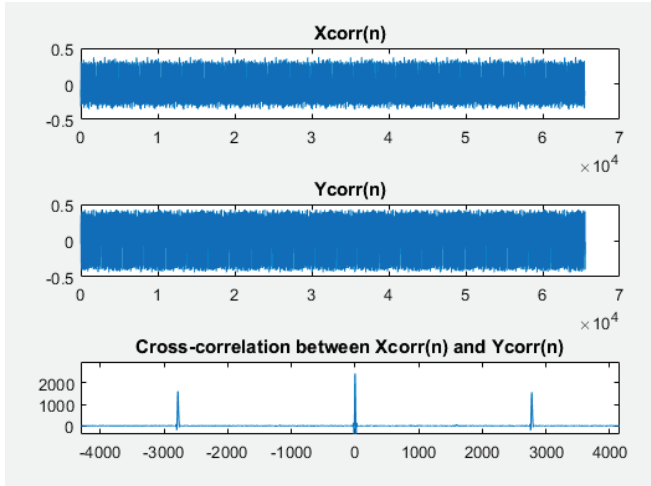


Fig. 3. After delay correction in the $x(n)$ and $y(n)$ signal

$Y_{corr}(n)$ is the PA output corresponding to input $X_{corr}(n)$.

B. Distortion Compensation Algorithms and Architectures [3]

In addition to modeling nonlinearities, the memory polynomial structure can be used as a pre-distorter (and post-distorter as well). The pre-distorter output Y is given by [6]

$$y_{MP}(n) = \sum_{k=0}^{K-1} \sum_{m=0}^{M-1} a_{km} x(n-m) |x(n-m)|^k$$

where x is the PA input, y is the PA output, a_{km} are the PA polynomial coefficients, M is the PA memory depth, K is the degree of PA non-linearity, n is the time index.

DPD is the process of find the inverse polynomial $f^{-1}(x_1, x_2, x_3, \dots, x_n)$ function of the PA model is

$$X_{corr_mk}(n) = \sum_{k=0}^{K-1} \sum_{m=0}^{M-1} p_{km} Y_{corr}(n-m) |Y_{corr}(n-m)|^k$$

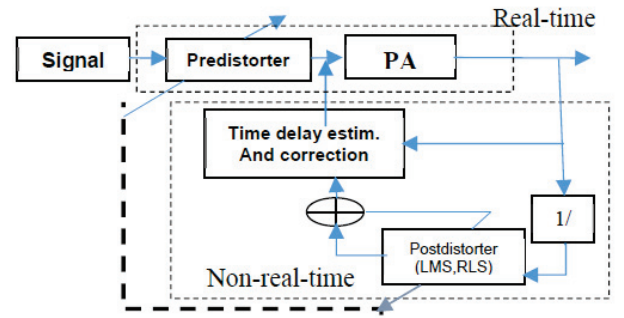


Fig. 4. Compensation using an indirect learning architecture

Indirect learning architecture (Figure 4) [4], which is a form of nonlinear adaptive inverse control, which is used for the adaptive identification of the pre-distorter. To obtain the inverse of the PA, the post-distorter uses an adaptive algorithm to adjust its parameters, which are the memory polynomial coefficients P_{km} in our case.

Solve for DPD coefficients, P_{km} . We re-write equation (10) as a set of a system of linear equations as shown below

$$\begin{pmatrix} X_{corr}(n) \\ X_{corr}(n+1) \\ \vdots \\ X_{corr}(n+p) \end{pmatrix} = \begin{pmatrix} Y_{corr}(n) & Y_{corr}(n-1) & \dots & Y_{corr}(n-m+1) & Y_{corr}(n-m+1)^2 & \dots \\ Y_{corr}(n+1) & Y_{corr}(n) & \dots & Y_{corr}(n-m+2) & Y_{corr}(n-m+2)^2 & \dots \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots \\ Y_{corr}(n+p) & Y_{corr}(n+p-1) & \dots & Y_{corr}(n-m+1+p) & Y_{corr}(n-m+1+p)^2 & \dots \end{pmatrix} \begin{pmatrix} P_{km} \\ P_{km} \\ \vdots \\ P_{km} \end{pmatrix}$$

The variable p denotes the number of measurement samples. The measurements x and y are known. A certain value for K and M are chosen based on the complexity of the PA.

Some Predistorter adaptation algorithms are[5]:

- Least Mean Square (LMS) Algorithm
- Normalized Least Mean Square (NLMS) Algorithm
- Recursive predictor error method (RPEM) Algorithm

Here, the complexity and implementation feasibility of the Pre-distorter adaptation algorithms, which is the LMS, is used.

III. IMPEMETATION

For advanced communication techniques, such as QPSK, OFDMA, and MIMO, it is very important to design an efficient and reliable system. In this paper Digital Pre-Distortion (DPD) technique is implemented for QPSK modulation scheme to reduce the adjacent channel power ratio (ACPR) which is used to measure the non-linearity in the adjacent channel and EVM for in-band distortion.

A. PA Characterization

PA is first characterized to understand its gain transfer function. The characterization setup is shown in Figure 5. Power amplifier gain is measured by continuous single tone signals fed from the signal generator for various input power and the PA output powers is measured.

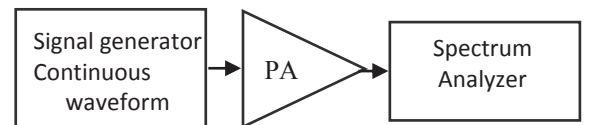


Fig. 5. PA Characterization setup

TABLE I. PA GAIN TRANSFER CHARACTERISTIC

Input Power (dBm)	Output Power (dBm)	Gain (dB)
-9	33.39	42.39
-8	34.34	42.34
-7	35.33	42.33
-6	36.27	42.27
-5	37.15	42.15
-4	38.00	42.00
-3	38.80	41.80
-2	39.60	41.60
-1	40.35	41.35
0	40.80	40.80
1	41.48	40.48
2	42.10	40.10
3	42.60	39.60
4	43.12	39.12
5	43.50	38.50
6	43.80	37.80
7	44.09	37.09
8	44.30	36.30
9	44.44	35.44
10	43.07	33.07

The transfer function and parameter extraction from the transfer characteristic is shown in Figure 6

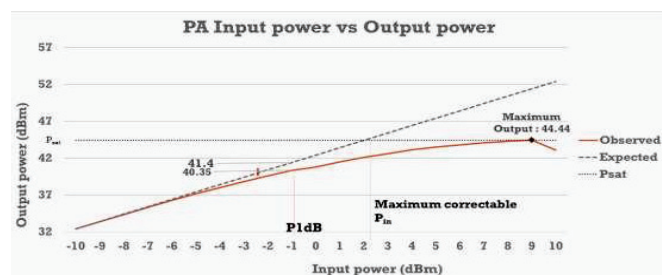


Fig. 6. Gain transfer function and PA Parameter extraction

Key parameters extracted from the Gain transfer characteristics are described in Table II

TABLE II. PA PARAMETERS

Parameter	Value
PA linear gain	42.3 dB
1 dB compression input power (P1dB)	-1 dBm
PA maximum output power (P _{sat})	44.44 dBm
Maximum correctable input power	2 dBm

Using time and delay estimation & correction described above, the following PA transfer function and inverse transfer function are analyzed

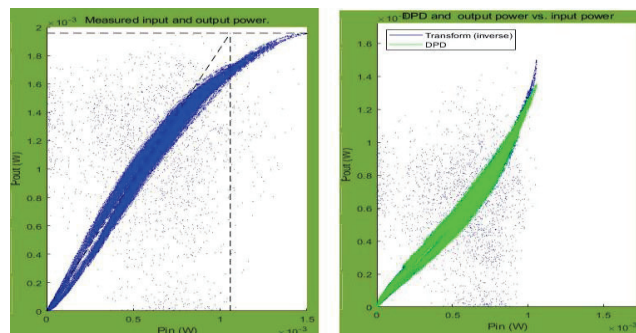


Fig. 7. PA Gain transfer function and Predistorter transfer function

B. Test Setup and DPD implantation

In this paper the Digital Pre-Distortion (DPD) technique using time & delay estimation and correction is implemented for QPSK/OFDM modulation schemes to reduce the adjacent channel power ratio (ACPR) which is used to measure the nonlinearity in the adjacent channel and EVM for in band distortion.

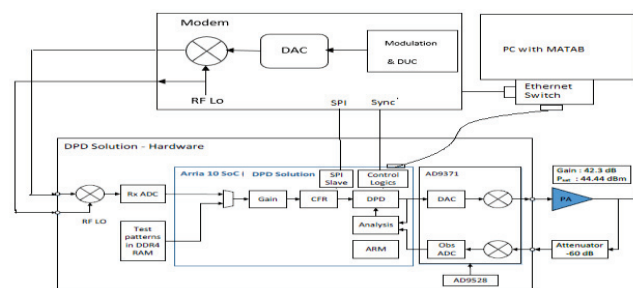


Fig. 8. Block diagram for Test setup

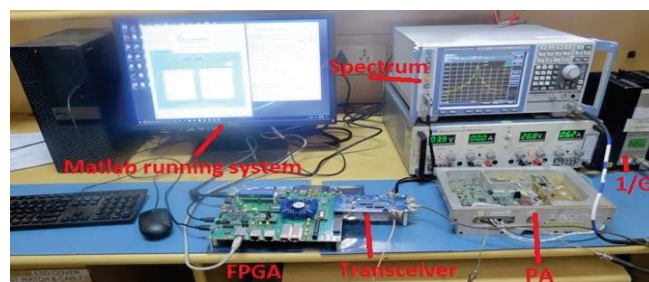


Fig. 9. Test setup for DPD in Lab

In the current setup, the DPD algorithm is implemented in Matlab. QPSK modulated signal is generated by modem and PA characteristic is analyzed in Matlab. Pre-distorted QPSK signal is sent through Ethernet to FPGA and upconverted and applied to a power amplifier and finally, output is observed in Spectrum Analyzer.

IV. RESULTS

QPSK modulated signal having ~ 15 MHz IBW is given to the Power amplifier having a 42.3dB gain and saturated power of 44.44dBm.

A. Matlab Simulation

ACPR and intermodulation distortion(IMD) simulation has been done in Matlab without and with DPD, and the results are shown in Fig.10 & Fig.11 respectively. With the implementation of DPD, there is an improvement of 31dB of ACPR. Improvement in phase error and magnitude error in the PA output with DPD is shown in Fig.12 as a constellation diagram

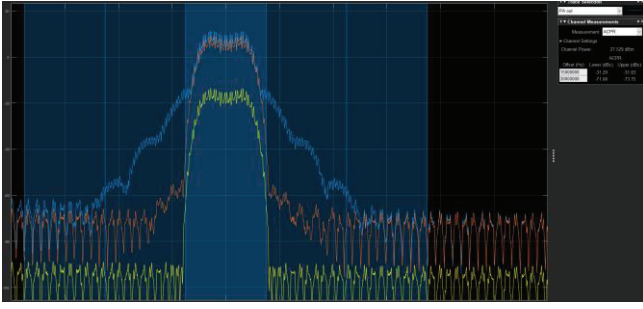


Fig. 10. Power Amplifier output without DPD (blue) and with DPD (orange). Measurement: APCR without DPD(-31.29dBc)

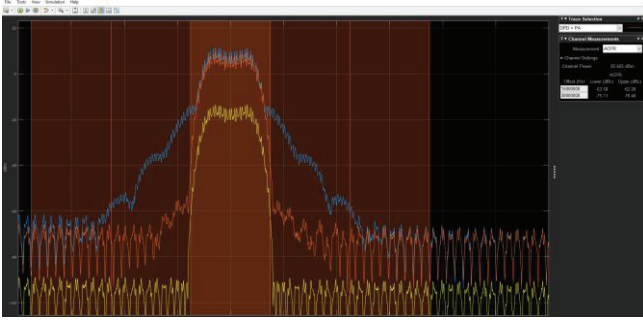


Fig. 11. Power Amplifier output without DPD (Blue) and with DPD (orange). Measurement: APCR with DPD (63.58dBc)

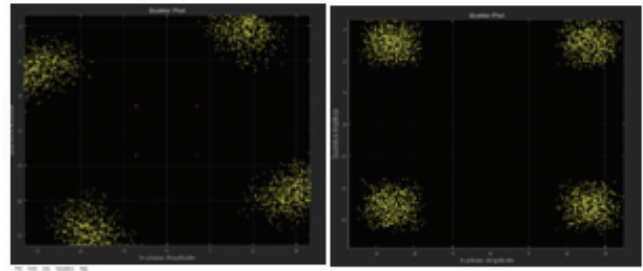


Fig. 12. Constellation Diagram of PA output for QPSK signal without DPD and with DPD

B. Result obtained in lab with DPD

DPD is tested using time & delay estimation and correction technique and found a significant improvement in the IMD rejection of PA. Figure 13 is PA output without DPD and figure 14 and 15 with DPD.



Fig. 13. Saturated PA output (No DPD)

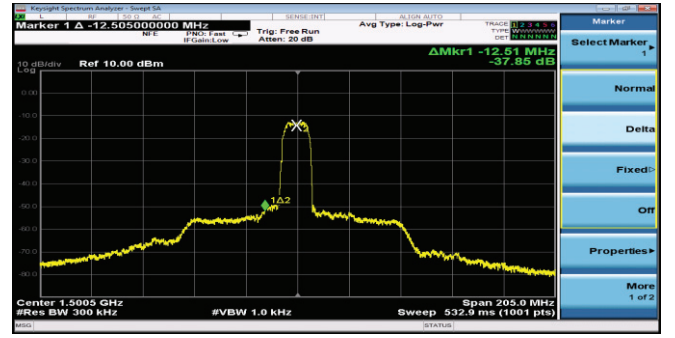


Fig. 14. Saturated PA output after DPD

There is 7.18 dB IMD improvement is found with DPD. Next time measurement was taken with time & delay and correction technique along with DPD. A significant improvement in IMD level is observed. Total 18.23 dB IMD rejection was obtained with time & delay and correction technique using correlation.

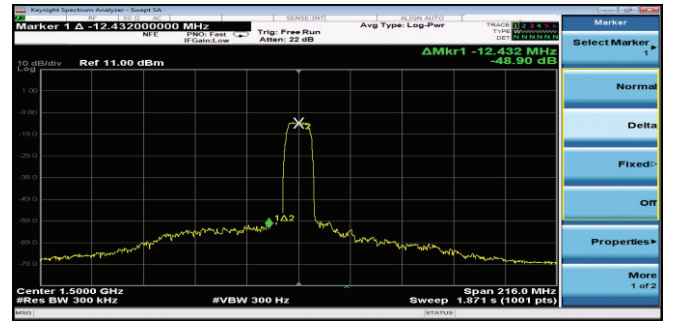


Fig. 15. Saturated PA output after DPD and time, delay and correction technique

TABLE III. COMPARISON TABLE OF PA NONLINEARITY

Test Conditions	Non-linearity (IMD level dBc)	Non-linearity improvement (dB)	Remarks
Saturated PA output (No DPD)	-30.67	---	
Saturated PA output after DPD	-37.85	7.18	With respect Saturated PA output
Saturated PA output after DPD and time, delay and correction technique	-48.90	18.23	With respect Saturated PA output

C. Result obtained for orthogonal frequency-division multiplexing (OFDM)

OFDM signal is also test with combining crest factor reduction (CFR) and digital predistortion and results show in figure 16 and 17.

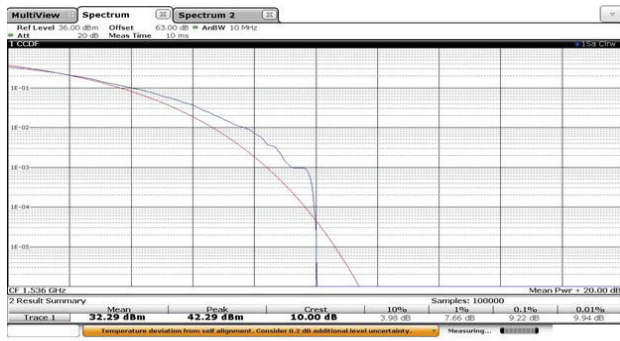


Fig. 16. CCDF of Power Amplifier Output without DPD

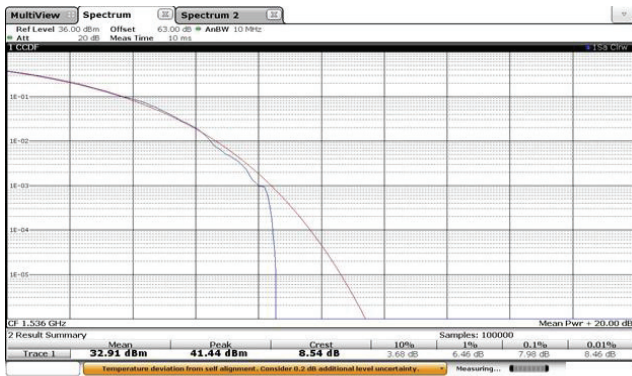


Fig. 17. CCDF of Power Amplifier Output with DPD

	Before DPD	After DPD
Crest Factor	10.00 dB	8.54 dB

V. CONCLUSION

This paper covers the problem and compensation of non-idealities of RF data links in the modern mobile communication system. The main issues and solutions

addressed were the nonlinear behavior of the HPA, dealt with using adaptive digital pre-distortion, and time delay estimation & correction techniques. There is an 11.05 dB improvement in IMD rejection by using the time delay estimation & correction technique. Matlab simulation and results obtained using the implementation of DPD in the lab are very similar to each other. Considering the application of the behavioral modeling approach to the RF link's nonlinear characterization was considered.

A hybrid system, which combines running the pre-distorter in Matlab and the adaptive training algorithms in aria 10 SoC based processor cards, was presented. The measurement results were obtained using the prototyping and demonstration system that was built so that nonlinear distortions can be substantially reduced by low complexity adaptive pre-distortion

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